Einstein when they met in Berlin in 1926 (as we saw in Chapter 2). And he was not alone—Pauli also counted relativity as a vindication of his godfather's views. The positivists viewed Einstein's work this way too. Moritz Schlick had first made his reputation as a philosopher with his book *Space and Time in Contemporary Physics,* a celebrated exposition of relativity and its philosophical implications. And the rest of the Vienna Circle was confident enough in Einstein's support of their ideas that it took the liberty of naming him one of the "leading representatives of the scientific world-conception" at the end of its manifesto.

Yet, despite the fact that he had borrowed from Mach, Einstein was not enamored of Mach's philosophy of science—at least, not later in life. "You know what I think about Mach's little horse," he wrote to a friend in 1919. "It cannot give birth to anything living. It can only exterminate harmful vermin." When Philipp Frank, a founding member of the Vienna Circle, asked Einstein about his philosophy of science, he was astonished to find that Einstein was not a positivist. Frank protested that Einstein had invented the positivist approach to physics in his theories of relativity. "A good joke should not be repeated too often," Einstein replied, much as he had to Heisenberg several years before.

Einstein certainly thought that science was about more than organizing perceptions. "What we call science," he said, "has the sole purpose of determining what *is.*" In an essay replying to Bohr and other critics in 1949, Einstein wrote that what he found unsatisfactory about quantum physics was that it denied the possibility of "the programmatic aim of all physics: the complete description of any (individual) real situation (as it supposedly exists irrespective of any act of observation or substantiation)." Einstein knew this was wildly out of step with the philosophical trends of the day—immediately after stating his beliefs about the aim of physics, he gave a sardonic aside about what he had just said, as seen from the point of view of an imagined positivist who takes both relativity and quantum theory as validation of his philosophical position:

> Whenever the positivistically inclined modern physicist hears such a formulation, his reaction is that of a pitying smile. He says to himself: "there we have the naked formulation of a metaphysical prejudice, empty of content, a prejudice, moreover, the conquest of which constitutes the major epistemological achievement of physicists within the last quarter-